



BLITTA – TOGO SOLAR POWER PLANT

Document Prepared by ELEMENT 6 - FZCO

ON BEHALF OF CLIMATEPARTNER GmbH



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1 PROJECT DETAILS

1.1 Summary Description of the Project

Togo, which imports more than half of its energy from Nigeria and Ghana, is banking on solar power to develop access to electricity for its eight million residents. The Togolese Government understands that the key to economic development and growth depends on access to electricity. The electrification strategy of Togo has set ambitious targets to achieve universal access by 2030.

The government of Togo has inaugurated the largest solar plant in West Africa and the first renewable energy facility in the country. This 70 MW solar PV facility, located in Blitta, will supply reliable, clean electricity to hundreds of thousands of homes and businesses in the country and save more than one million tonnes of CO₂ emissions through its lifetime.

The village of Blitta-Losso is part of the canton of Blitta village and is surrounded from north to south and from east to west by the villages of Blitta kotokoli, Boufouli boko losso, Doufouli and Oranyi and extends over the parallels Latitude: 8°20' latitude and 1°01' longitude (Geographical coordinates in degrees minutes seconds (WGS84).

The Blitta area has an average monthly direct normal radiation of 5.14 kWh/m²/day and an average annual solar irradiation of 1,876 kWh/m².

The project will advance the Togolese national clean energy strategy to increase the share of renewables in electrification to 50 per cent by 2025 and 100 per cent by 2030.

The project activity, which has a lifetime of 25 years, is expected to produce more than 88,000 MWh of electricity annually. The total estimated emission reductions of the proposed project are 464,000 tCO₂e during the first crediting period. The ex-ante emission reduction calculation shows that the project will avoid 66,000 tCO₂e on average, annually.

The project activity is the installation of a new power plant at the site and is not a capacity addition or retrofit or replacement of any other existing plant. Any electricity generated and exported from the project activity would in its absence have been generated in the grid dominated by fossil fueled power plants, which represents the baseline scenario. The project scenario is a Solar photovoltaic power plant that would generate energy and supply it to the national grid through La Compagnie Energie Electrique du Togo.

The project will not only reduce the GHGs emissions but is expected to achieve many social benefits such as improving social services to targeted social groups and generating job opportunities.

1.2 Sectoral Scope and Project Type

Sectoral Scope: 1, Energy industries (renewable -/non-renewable sources)

Type: Renewable energy projects (Large Scale)

Category: ACM 0002 - Large-scale Consolidated Methodology, Grid-connected electricity generation from renewable sources (Version 20.0)

The methodology refers to ACM0002 to the latest approved version of following tools

- “TOOL01: Tool for the demonstration and assessment of additionality” version 7.0.0
- “TOOL07: Tool to calculate the emission factor for an electricity system” version 7.0
- “TOOL23: Additionality of first-of-its-kind project activities” version 3.0

The project is a renewable energy type and is not a grouped project.

1.3 Project Eligibility

According to VCS Standard, version 4.2, the project is eligible under the scope of the VCS Program because:

- It results in CO₂ emission reductions, one of the six Kyoto Protocol greenhouse gases
- It consists of grid-connected electricity generation using (large scale) solar power plant built in a Least Developed Country.

1.4 Project Design

The project has been designed to include a single installation of an activity.

- ☒ The project includes a single location or installation only
- ☐ The project includes multiple locations or project activity instances, but is not being developed as a grouped project
- ☐ The project is a grouped project

Eligibility Criteria

The proposed project is not a group project activity. Thus, this section is not applicable.

For grouped projects, provide additional information relevant to the design of the grouped project (e.g., the eligibility criteria for the inclusion of new project activity instances).

1.5 Project Proponent

Organization name	AMEA Power LLC
Contact person	Basma Bentaher
Title	Senior Director – Project Development
Address	Marina Plaza Offices 3301/3302/3303 Level 33 Dubai Marina PO Box 37669 Dubai United Arab Emirates
Telephone	+971 4 310 7014
Email	Basma.Bentaher@ameapower.com

Organization name	ClimatePartner GmbH
Contact person	Dr. Sascha Lafeld
Title	Head of Carbon Offsets and Green Energy Services
Address	St.-Martin-Str. 59, D-81669 Munich, Germany
Telephone	+49 89 1222 8750
Email	sascha.lafeld@climatepartner.com

1.6 Other Entities Involved in the Project

Organization name	Element 6 – FZCO
Role in the project	Consultant
Contact person	Amer Arafat
Title	Partner
Address	Technohub 1, DTEC, Dubai Silicon Oasis, Dubai, UAE
Telephone	+971 52 938 9590
Email	amer@element6.cc

1.7 Ownership

The project activity is fully owned by the project proponent AMEA Power LLC. The ownership of the project can be proven through:

- Certificate of Completion signed by the Togolese Republic
- Power Purchase Agreement (PPA) signed by AMEA Power LLC and the Togolese Republic
- Declaration of Conformity signed by the Togolese Republic
- Entry into Force of the Concession Agreement signed by AMEA Power LLC and the Togolese Republic
- Back Energization of the PV Plant Document signed by the Togolese Republic, the Electricity Off-taker and AMEA Power LLC.

1.8 Project Start Date

The project's COD (Commercial Operation Date) was 12 Jun 2021, which is date on which the project started generating emission reductions.

1.9 Project Crediting Period

The project's crediting period is 7 years renewable.

The first crediting period starts from 12 Jun 2021 and lasts till 11 Jun 2028.

1.10 Project Scale and Estimated GHG Emission Reductions or Removals

The estimated annual GHG emission reductions/removals of the project are:

- ☐ <20,000 tCO₂e/year
- ☒ 20,000 – 100,000 tCO₂e/year
- ☐ 100,001 – 1,000,000 tCO₂e/year
- ☐ >1,000,000 tCO₂e/year

Project Scale	
Project	√
Large project	

Year	Estimated GHG emission reductions or removals (tCO ₂ e)
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2021	25,017
2022	50,496
2023	70,700
2024	70,700
2025	70,700
2026	70,700
2027	70,700
2028	35,027
Total estimated ERs	464,040
Total number of crediting years	7
Average annual ERs	66,291

1.11 Description of the Project Activity

The proposed project activity consists of setting up a 70 MW solar PV plant in Blitta-Lasso in Togo. The land is 117 ha large and is made available by the Togolese Republic. The Blitta area has an average monthly direct normal radiation of 5.14 kWh/m²/day and an average annual solar irradiation of 1,876 kWh/m².

Togo recognizes that access to electricity is an essential element for its economic development and inclusive growth policy and is one of the major pillars that will enable the implementation of its new National Development Plan. Their goal is to ensure universal access for all Togolese by 2030, with a 100% access rate over the next ten years and will be achieved through an intelligent combination of network expansion and off-grid technologies (mini-grid solar kits).

Table 1 - Characteristics of the PV Panels

Manufacturer	Jinko Solar
Model	390 Wp (JKM-390M-72V) & 395 Wp (JKM-395-72V) 400 Wp (JKM-400M-72V) single crystal
Power installed on the ground	50 MWp with 20 MWp planned expansion
Number of panels installed	127,344
Total panel area	252,684 m ²

Table 2 - Inverter Specifications

Make and model	Sungrow SG250HX- 250KTL – H1 String or equivalent
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Nominal apparent power	250 kVA @30°C/ 225 kVA @40°C/200 kVA @50°C
Rated current (In)	126.3A @40°C, 121.3A @45°C, 108.3A @50°C
Maximum apparent power	200 × 250,000 VA
Rated output voltage	800 V
Type of connection	Triphase

Table 3 - Characteristics of the inverter flow transformers

Model	External mineral oil (STS-6000K) or equivalent
Rated power	6 × 6000 kVA @40°C/5400 kVA @50°C and 2 x 8000 kVA @40°C
Primary voltage	800 V
Secondary voltage	33 kV
No-load losses	6.25% (±10%)
Losses at rated current	<4.8 kW (IEC 60076-1) (IEC 62271-202, EN 50588-1, IEC 61429-1)

1.12 Project Location

The Government of Togo (Ministry of Mining and Energy) signed a Memoranda of Understanding with the solar developer, AMEA Power LLC, aiming to develop a 70 MW solar power plant in Togo. The coordinates of the project location are 8°21'52.33"N, 1° 1'4.42"E and covers a land area of 117ha. Additionally, the PV plant will be connected to the national grids 161 kV transmission line located close to the site. The Blitta area has an average monthly direct normal radiation of 5.14 kWh/m²/day and an average annual solar irradiation of 1,876 kWh/m².

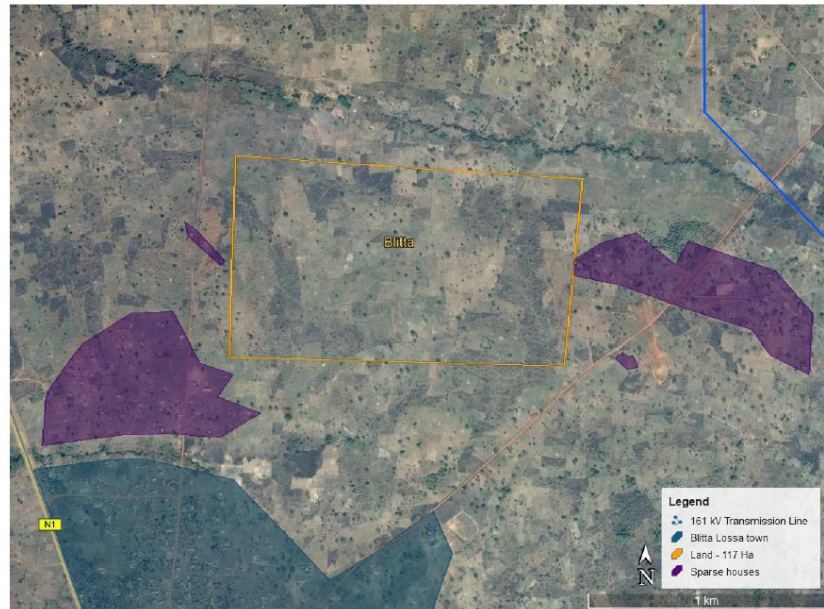


Figure 1 - Site location for the 70MW PV Power Plant

1.13 Conditions Prior to Project Initiation

According to ACM0002 Version 20 and since the project activity is the installation of a new grid-connected renewable power plant (Greenfield), the baseline scenario is the following:

“If the project activity is the installation of Greenfield power plant, the baseline scenario is electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in the “Tool to calculate the emission factor for an electricity system.”

1.14 Compliance with Laws, Statutes and Other Regulatory Frameworks

The Togolese government created the Togolese Rural Electrification and Renewable Energies Agency (AT2ER), whose mission is to promote renewable energy in Togo. The creation of the AT2ER was accompanied in July 2018 by the announcement of the Law 2018-010 08/08/2018 on the promotion of electricity production based on renewable energy sources in Togo. This resulted in creation of a specific legal framework and incentives to respond adequately to the challenges related to the production, marketing and use of electricity generated from renewable energy in Togo.

1.15 Participation under Other GHG Programs

1.15.1 Projects Registered (or seeking registration) under Other GHG Program(s)

The project is neither registered nor seeking registration under any other GHG programs.

1.15.2 Projects Rejected by Other GHG Programs

The project has not been rejected by any other GHG programs.

1.16 Other Forms of Credit

1.16.1 Emissions Trading Programs and Other Binding Limits

Does the project reduce GHG emissions from activities that are included in an emissions trading program or any other mechanism that includes GHG allowance trading?

☐ Yes ☒ No.

1.16.2 Other Forms of Environmental Credit

Has the project sought or received another form of GHG-related credit, including renewable energy certificates?

☐ Yes ☒ No

1.17 Sustainable Development Contributions

In 2018, only 45% of Togolese population had access to electricity, which included only 8% in rural areas. In order to meet the needs of the Togolese people, the Government of Togo has committed to an ambitious policy on energy access that is to achieve a 90% electricity access rate in the country by 2028. In this context, the country has subscribed at regional and international level to many commitments in this sector and which require that the share of renewable energy to be increased to 50% in the national energy mix by 2030. This is in accordance with Goal 7 of the United Nations Sustainable Development Goals, which aims to ensure access for all to reliable, sustainable, and modern energy services and at affordable costs.

It is also important to highlight the substantial work that AMEA Power has conducted over on the social aspect as well that meet several SDGs:

- 1) Constructed a Medical Clinic for the village of Blitta
- 2) Constructed 3 primary schools for the neighboring villages
- 3) Renovated an additional 4 primary schools in other villages including furniture, boards, stationery etc...
- 4) Installed a water pump to provide potable water for the local primary school at Blitta
- 5) Ensured the schools were provided with free electricity from the Solar Power Plant
- 6) Conducted a comprehensive internship program for almost 40 students from the Lomé Technical University, providing accommodation and transport
- 7) Conducted an essential food and hygiene product program to over 2,200 families for those in need during the country lockdown response to COVID-19

1.18 Additional Information Relevant to the Project

Leakage Management

As per approved methodology ACM0002 (Version 20.0), no leakage is considered for the proposed project activity.

Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the project description.

Further Information

N/A

2 SAFEGUARDS

2.1 No Net Harm

The project activities during construction, operation, and decommissioning of the project have been analysed in order to identify corresponding environmental and social impacts and assess their significance. Mitigation measures have been identified in the Environmental and Social Management Plan (ESMP) for negative impacts where institutional responsibilities for implementation and follow up have been presented. The socioeconomic benefits of the proposed project therefore outweigh the potential negative environmental and social impacts.

2.2 Local Stakeholder Consultation

Describe the process for, and the outcomes from, the local stakeholder consultation conducted prior to validation. Include details on the following:

- *The procedures or methods used for engaging local stakeholders (e.g., dates of announcements or meetings, periods during which input was sought).*
- *The procedures or methods used for documenting the outcomes of the local stakeholder consultation.*
- *The mechanism for on-going communication with local stakeholders.*
- *How due account of all and any input received during the consultation has been taken. Include details on any updates to the project design or justify why updates are not appropriate.*

2.3 Environmental Impact

An Environmental and Social Impact Assessment (ESIA) was carried out for the project. The project activities during construction and operation of the project have been analyzed to identify correspondent environmental impacts and assess their significance. The summary of the findings of the report is as follows:

Impact Assessment methodology

The following assessment of impacts distinguishes between the construction, operation, and decommissioning phases.

- Land use
- Biodiversity
- Air quality and noise levels
- Water resources
- Traffic and Transport
- Existing infrastructure
- Visual impacts
- Socioeconomic impacts

Land, land use, and waste management

As the land consists of trees and agricultural activities, there might be some changes in the soil properties resulting from the soil replacement for the foundations and from construction of the plant intermediate road. Such impacts could only affect the subsequent use of the project site after decommissioning and would be mitigated through the site preparation works. There might also be some impacts on the soil at the project site resulting from accidental leaks of wastes or oils used by the construction machinery. However, such spills are expected to be limited through the Spill Prevention Plans and having spill kits available throughout the site and the Hazardous Waste Management plan.

Impacts on land beyond the footprint of the project will also include domestic solid waste in addition to some construction waste such as plastics, metals, wood, etc. Because the quantity of such a waste was expected to be limited, the correspondent impacts on waste disposal sites would be minor.

Air quality

The air emissions expected during the construction of the project were the airborne dust resulting from earthworks, in addition to the emissions of different construction machinery such as power generators, excavators, loaders, vehicles, etc. In the absence of sensitive receptors in the area, emissions during construction, such as exhaust gases have low significant impact, and may only

temporarily affect users, as the average speed on the road is quite high, during passing the area in addition to slight impacts on the construction workers.

Mitigation measures such as spraying loose soil during excavation and filling operation can be adopted to minimize airborne dust.

Noise levels

Normally construction works include noisy activities related to the operation of construction machines, possible hammering, drilling works, soil compaction in addition to the noise generated from construction related trucks. The impacts of construction noise are likely to cause some considerable effects on neighbors as the neighboring areas are closely populated.

The contractor was encouraged to minimize noise emissions, such as turning off engines of unused equipment, use lower noise emission equipment to the extent possible and use silencer for noisy equipment (such as generators) to comply with the local noise regulations.

Water resources

During construction, temporary movable construction yard (for storage of materials and servicing of machinery) and temporary offices, for construction staff, were erected at a place. The area would include simple sanitary facilities. Water supply was via tankers from the central pipeline, which will normally cover domestic uses, uses for soil spraying and uses for mixing concrete. The amount of water consumed in construction sites are expected to be relatively little amounts, in which the quantitative impacts on water resources are negligible. The mitigation measures include collecting wastewater in waterproof collection tanks and evacuate it frequently to the nearest wastewater treatment plant or sewerage system.

Traffic and Transport

During the construction, heavy transport was used to transfer PV modules and other electric equipment to the construction sites. Since the road networks have low levels of service, truck convoys carrying equipment to the site may cause bottlenecks and impact the current traffic flow.

A plan to develop traffic management was introduced to ensure equipment is delivered in a timely fashion and in good condition, whilst complying with the local legislation.

2.4 Public Comments

Demonstrate how due account of all and any comments received during the public comment period has been taken. Include details on any updates to the project design or demonstrate the insignificance or irrelevance of comments.

To be updated

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

The methodology applied is:

- ACM0002 “Large-scale Consolidated Methodology - Grid-connected electricity generation from renewable sources, version 20.0”.

The methodology refers to the following methodological tools:

- TOOL1 “Tool for the demonstration and assessment of additionality” version 7.0.0
- TOOL7 “Tool to calculate the emission factor for an electricity system” version 7.0
- TOOL23 “Additionality of first-of-its-kind project activities” version 3.0
- TOOL32 “Positive lists of technologies” version 4.0

3.2 Applicability of Methodology

The project activity is a solar project with a total installed capacity of 70 MW. As such, “Large-scale Consolidated Methodology ACM0002 - Grid-connected electricity generation from renewable sources, version 20.0” is applicable for this project. The project activity involves the installation of a new power plant at a site where no renewable power plant previously existed and as such, is a Greenfield Power Plant.

The methodology is applicable under the following conditions:

Methodological Applicability	Project Justification
This methodology is applicable to grid-connected renewable energy power generation project activities that: (a) Install a Greenfield power plant; (b) Involve a capacity addition to (an) existing plant(s); (c) Involve a retrofit of (an) existing operating plants/units; (d) Involve a rehabilitation of (an) existing plant(s)/unit(s);or (e) Involve a replacement of (an) existing plant(s)/unit(s).	Option (a) - The project is a grid-connected solar energy power generation project that is a greenfield installation.

The methodology is applicable under the following conditions: a) The project activity may include renewable energy power plant/unit of one of the following types: hydropower plant/unit with or without reservoir, wind power plant/unit, geothermal power plant/unit, solar power plant/unit, wave power plant/unit or tidal power plant/unit; b) In the case of capacity additions, retrofits, rehabilitations or replacements (except for wind, solar, wave or tidal power capacity addition projects) the existing plant/unit started commercial operation prior to the start of a minimum historical reference period of five years, used for the calculation of baseline emissions and defined in the baseline emission section, and no capacity expansion, retrofit, or rehabilitation of the plant/unit has been undertaken between the start of this minimum historical reference period and the implementation of the project activity.	Option (a) – The project activity is a renewable energy (solar power) power plant.
In case of hydro power plants, one of the following conditions shall apply: a) The project activity is implemented in existing single or multiple reservoirs, with no change in the volume of any of the reservoirs; or b) The project activity is implemented in existing single or multiple reservoirs, where the volume of the reservoir(s) is increased and the power density, calculated using equation (7), is greater than 4 W/m²; or	Not applicable. This project is not a hydro power plant.

c) The project activity results in new single or multiple reservoirs and the power density, calculated using equation (7), is greater than $4W/m^2$; or d) The project activity is an integrated hydro power project involving multiple reservoirs, where the power density for any of the reservoirs, calculated using equation (7), is lower than or equal to $4W/m^2$, all of the following conditions shall apply: i. The power density calculated using the total installed capacity of the integrated project, as per equation (8), is greater than $4 W/m^2$; ii. Water flow between reservoirs is not used by any other hydropower unit which is not a part of the project activity; iii. Installed capacity of the power plant(s) with power density lower than or equal to $4W/m^2$ shall be: a) Lower than or equal to 15MW; and b) Less than 10 per cent of the total installed capacity of integrated hydro power project.	
In the case of integrated hydro power projects, project proponent shall: a) Demonstrate that water flow from upstream power plants/units spill directly to the downstream reservoir	Not applicable. This project is not an integrated hydro power plant.

and that collectively constitute to the generation capacity of the integrated hydro power project; or b) Provide an analysis of the water balance covering the water fed to power units, with all possible combinations of reservoirs and without the construction of reservoirs. The purpose of water balance is to demonstrate the requirement of specific combination of reservoirs constructed under CDM project activity for the optimization of power output. This demonstration has to be carried out in the specific scenario of water availability in different seasons to optimize the water flow at the inlet of power units. Therefore, this water balance will take into account seasonal flows from river, tributaries (if any), and rainfall for minimum of five years prior to the implementation of the CDM project activity.	
The methodology is not applicable to: a) Project activities that involve switching from fossil fuels to renewable energy sources at the site of the project activity, since in this case the baseline may be the continued use of fossil fuels at the site; b) Biomass fired power plants/units.	Not applicable. This project activity does not involve switching of fossil fuels to renewable energy or biomass fired power plants.
In the case of retrofits, rehabilitations, replacements, or capacity additions, this methodology is only applicable if the most plausible baseline scenario, as a result of the identification of baseline scenario, is “the continuation of the current situation, that is to use the power generation equipment that was already in use prior to the implementation of	Not applicable. This project activity is not a retrofit, rehabilitation, replacement, or capacity addition.

the project activity and undertaking business as usual maintenance”.	
In addition, the applicability conditions included in the tools referred to below apply.	See below tables.

TOOL7 “Tool to calculate the emission factor for an electricity system” version 7.0 is applicable under the following conditions:

Tool Applicability	Project Justification
This tool may be applied to estimate the OM, BM and/or CM when calculating baseline emissions for a project activity that substitutes grid electricity that is where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g. demand-side energy efficiency projects).	The project is a grid-connected solar energy power generation project and thus substitutes grid electricity.
Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an option, can include off-grid power plants. In the latter case, two sub-options under the step 2 of the tool are available to the project participants, i.e. option IIa and option IIb. If option IIa is chosen, the conditions specified in “Appendix 1: Procedures related to off-grid power generation” should be met. Namely, the total capacity of off-grid power plants (in MW) should be at least 10 per cent of the total capacity of grid power plants in the electricity system; or the total electricity generation by off-grid power plants (in MWh) should be at least 10 per cent of the total electricity generation by grid power plants in the electricity system; and that factors which negatively affect the reliability and stability of the grid are primarily due to constraints in generation and not to other aspects such as transmission capacity.	Steps involved in calculation of Emission Factor is included in section 3.1 of the VCS PD as per the requirement of the tool.

In case of CDM projects the tool is not applicable if the project electricity system is located partially or totally in an Annex I country.	Project is located in non-Annex I country and hence the tool is applicable.
Under this tool, the value applied to the CO2 emission factor of biofuels is zero.	Not applicable. No biofuel is involved in this project activity.

3.3 Project Boundary

The spatial extent of the project boundary includes the project power plant/unit and all power plants/units connected physically to the electricity system that the solar power plant is connected to.

The electricity system is defined by the spatial extent of the power plants that are physically connected through transmission and distribution lines to the solar power plant and that are covered by either single or layered dispatch area.

Thus, the project boundary for this activity is the national/sub-national grid in Togo.

Source	Gas	Included?	Justification/Explanation
Baseline	CO ₂	Included	Main emission source
	CH ₄	Excluded	Minor emission source
	N ₂ O	Excluded	Minor emission source
Project	CO ₂	Included	Main emission source
	CH ₄	Excluded	Minor emission source
	N ₂ O	Excluded	Minor emission source

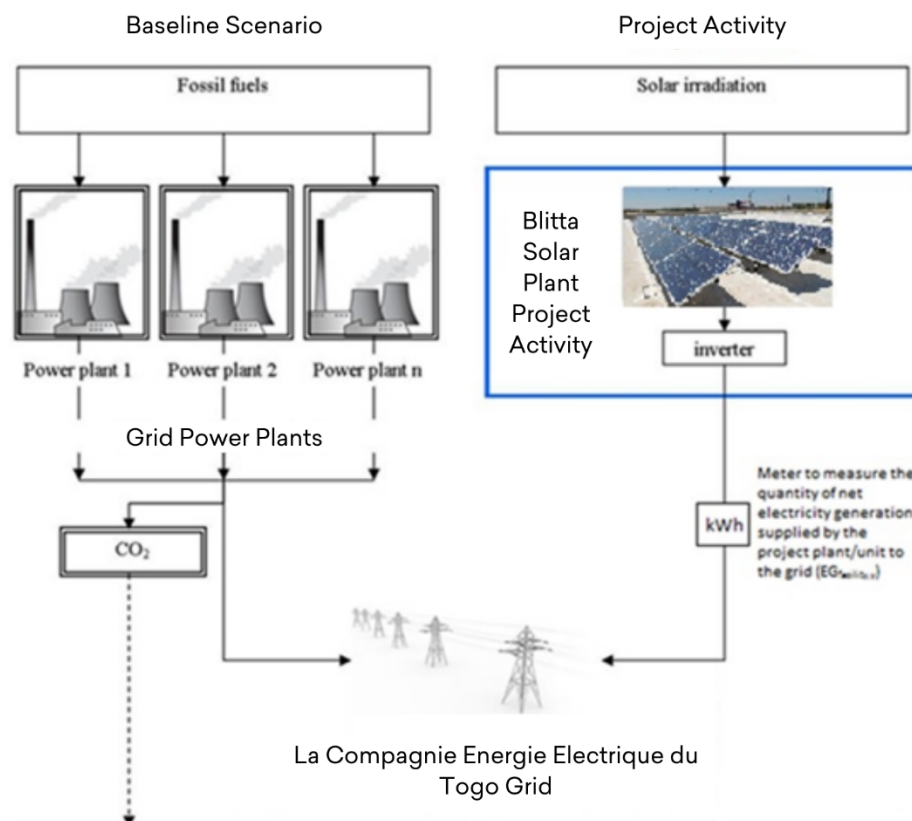


Figure 2 - Diagram of project boundary

3.4 Baseline Scenario

According to the ACM0002 Large Scale Consolidated Methodology:

“If the project activity is the installation of a Greenfield power plant, the baseline scenario is electricity delivered to the grid by the project activity that would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in “TOOL07: Tool to calculate the emission factor for an electricity system.”

The project activity is the installation of a new power plant at the site and is not a capacity addition or retrofit or replacement of any other existing plant. Any electricity generated and exported from the project activity would in its absence have been generated in the grid dominated by fossil fueled power plants, which represents the baseline scenario. The project scenario is a Solar photovoltaic power plant that would generate energy and supply it to the national grid through La Compagnie Energie Electrique du Togo.

The baseline, the scenario existing prior to the implementation of the project activity, is the continuation of the current practice. The electricity generated by the project activity would have been generated by the addition of new fossil fuel-based generation sources to the grid to supply the additional electricity capacity required by demand.

3.5 Additionality

According to Large-scale Consolidated Methodology ACM0002 - Grid-connected electricity generation from renewable sources, version 20.0", section 5.3.1, the simplified procedure to demonstrate additionality can be used, using TOOL32 "Positive lists of technologies".

Hence, the project activity is covered in the positive list. Additional to this, according to TOOL1 "Tool for the demonstration and assessment of additionality" version 7.0.0, section 4.1, if the proposed project activity is the first-of-its-kind, its additionality is demonstrated.

Using TOOL23 "Additionality of first-of-its-kind project activities", section 5.1., paragraph 12, the proposed project activity is identified as first-of-its-kind in the applicable geographical area because the following conditions are met:

Methodology Procedure	Justification
a) The project is the first in the applicable geographical area that applies a technology that is different from technologies that are implemented by any other project, which are able to deliver the same output and have started commercial operation in the applicable geographical area before the project design document (CDM-PDD) is published for global stakeholder consultation or before the start date of the proposed project activity, whichever is earlier;	The project is the first utility-scale solar power plant in Togo.
b) The project implements one or more of the measures	As per paragraph 9 of TOOL23, one of the four types of measures currently covered in the framework is: b) Switch of technology with or without change of energy source including energy efficiency improvement as well as use of renewable energies (example: energy

	efficiency improvements, power generation based on renewable energy); This renewable energy power generation project therefore implements one of the measures.
c) The project participants selected a crediting period for the project activity that is “a maximum of 10 years with no option of renewal”.	The project is being registered for a crediting period of 10 years, non-renewable.

The conclusion is that the project activity is indeed first-of-its-kind and hence is additional.

3.6 Methodology Deviations

Not applicable – there are no deviations.

4 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

According to the latest version of the methodology, ACM0002 Large Scale Consolidated Methodology: Grid-connected Electricity Generation from Renewable Sources, Version 20.0, since the project activity is the installation of Greenfield Renewable Energy Power Plant, the baseline scenario is electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in “TOOL07: Tool to calculate the emission factor for an electricity system”.

The scope and applicability of Standardized Baseline ASB0034-2021: Grid emission factor for the West African Power Pool, covers the grid emission factors for the electricity system of the WAPP (West African Power Pool).

This Standardized Baseline can be used under the following conditions:

Baseline Procedure	Justification
The project activity is implemented in any one of the WAPP member	The project activity is in <u>Togo</u> and is connected to the WAPP.

countries i.e. Benin, Burkina Faso, Côte d'Ivoire, Ghana, Mali, Niger, Nigeria, Senegal and <u>Togo</u> and is connected to the WAPP	
The CDM approved methodology that is applied to the project activity requires the determination of CO ₂ emission factor(s) through the application of the grid tool	ACM0002 requires the determination of CO ₂ emission factor(s) through the application of the grid tool "TOOL07: Tool to calculate the emission factor for an electricity system
The project activity uses the ex-ante options for both the operating margin and build margin grid emissions factors, as described in the grid tool, and therefore no monitoring or recalculation of the emission factor during the crediting period is required	The project activity uses the ex-ante options for both the operating margin and build margin grid emission factors, as described in the grid tool.

According to the ACM0002, Baseline emissions include only CO₂ emissions from electricity generation in fossil fuel-fired power plants that are displaced due to the project activity. The methodology assumes that all project electricity generation above the baseline levels would have been generated by existing grid-connected power plants and the addition of new grid-connected power plants.

The baseline emissions are to be calculated as follows:

$BE_y = EG_{PJ,y} \times EF_{grid,CM,y}$ Where:

BE_y = Baseline emissions in year y (tCO₂/yr)

$EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh/yr)

$EF_{grid,CM,y}$ = Combined margin CO₂ emission factor for grid connected power generation in year y calculated using the latest version of "TOOL07: Tool to calculate the emission factor for an electricity system" (tCO₂/MWh)

Calculation of $EG_{PJ,y}$

In case of Greenfield power plant development, the calculation of the electricity system's combined margin (CM) shall be based on the following equation, according to the methodology:

$EG_{PJ,y} = EG_{facility,y}$ Where:

$EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh/yr)

$EG_{facility,y}$ = Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh/yr)

Calculation of $EF_{grid,CM,y}$

As stated before, the “TOOL07: Tool to calculate the emission factor for an electricity system” allows referring to the Standardized Baseline ASB0034-2021: Grid emission factor for the West African Power Pool.

The ASB0034-2021 Standardized Baseline states that for wind and solar power generation project activities, the $EF_{grid,CM,y}$ to be used is 0.573 tCO₂/MWh.

4.2 Project Emissions

The project activity involves renewable power generation. Therefore, $PE_y = 0$

4.3 Leakage

No leakage emissions are considered. The emissions potentially arising due to activities such as power plant construction and upstream emissions from fossil fuel use (e.g., extraction, processing, transport etc.) are not calculated.

4.4 Net GHG Emission Reductions and Removals

Baseline Emissions:

Baseline Emissions are calculated as follows:

$$BE_y = EG_{PJ,y} * EF_{grid,CM,y}$$

Where:

BE_y = Baseline emissions in year y (tCO₂ /year)

$EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the VCS project activity in year y (MWh/yr)

$EF_{grid,CM,y}$ = Combined margin CO₂ emission factor for grid connected power generation in year y. The value is assumed based on the Standardized Baseline ASB0034-2021: Grid emission factor for the West African Power Pool calculated using the latest version of the “Tool to calculate the emission factor for an electricity system” (tCO₂/MWh).

The quantity of net electricity supplied to the grid in phase 1 is:

$$EG_{pj,y} = 88,126 \text{ MWh/yr}$$

The quantity of net electricity supplied to the grid when phases 1 and 2 are operational:

$$EG_{pj,y} = 123,386 \text{ MWh/yr}$$

The Grid Emission Factor is:

$$EF_{grid,CM,y} = 0.573 \text{ tCO}_2/\text{MWh}$$

The calculation of the baseline emissions of phase 1 (50 MW) of the proposed project activity is:

$$BE_y = 88,126 \text{ MWh/yr} * 0.573 \text{ tCO}_2/\text{MWh} = 50,496 \text{ tCO}_2/\text{yr}$$

The calculation of the baseline emissions of phases 1 and 2 (70 MW) of the proposed project activity is:

$$BE_y = 123,386 \text{ MWh/yr} * 0.573 \text{ tCO}_2/\text{MWh} = 70,700 \text{ tCO}_2/\text{yr}.$$

Leakage:

Leakage emissions are not applicable.

Project Emissions:

Project Emissions are as shown below:

$$PE_y = 0 \text{ tCO}_2/\text{yr}$$

Emission Reductions:

$$ER_y = BE_y - PE_y$$

$$ER_y \text{ (Phase 1)} = 50,496 \text{ tCO}_2/\text{yr} - 0 \text{ tCO}_2/\text{yr} = 50,496 \text{ tCO}_2/\text{yr}$$

$$ER_y \text{ (Phases 1 and 2)} = 70,700 \text{ tCO}_2/\text{yr} - 0 \text{ tCO}_2/\text{yr} = 70,700 \text{ tCO}_2/\text{yr}$$

Year	Estimated baseline emissions or removals (tCO ₂ e)	Estimated project emissions or removals (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated net GHG emission reductions or removals (tCO ₂ e)
Year 2021	25,017	0	0	25,017
Year 2022	50,496	0	0	50,496
Year 2023	70,700	0	0	70,700
Year 2024	70,700	0	0	70,700

Year 2025	70,700	0	0	70,700
Year 2026	70,700	0	0	70,700
Year 2027	70,700	0	0	70,700
Year 2028	35,027	0	0	35,027
Total	464,040	0	0	464,040

5 MONITORING

5.1 Data and Parameters Available at Validation

Data / Parameter	EF grid, CM, y
Data unit	tCO ₂ /MWh
Description	Combined Margin CO ₂ grid emission factor in the year y
Source of data	Standardized Baseline ASB0034-2021: Grid emission factor for the West African Power Pool
Value applied	0.573 tCO ₂ /MWh
Justification of choice of data or description of measurement methods and procedures applied	XXX
Purpose of Data	<i>Calculation of baseline emissions</i>
Comments	This parameter is fixed ex-ante for the entire crediting period

5.2 Data and Parameters Monitored

Data / Parameter	EG PJ,y
Data unit	MWh/yr
Description	Quantity of net electricity generation supplied to the grid in year y by the solar plant
Source of data	<i>Meter readings</i>
Description of measurement methods	Net electricity generation will be electronically metered and stored by XXX under the Plant Manager's authority

and procedures to be applied	
Frequency of monitoring/recording	<i>Continuous measurement and monthly recordings</i>
Value applied	XXX
Monitoring equipment	XXX
QA/QC procedures to be applied	Calibration of the meter will be carried out as per manufacturer specifications. Annual inspection and testing of accuracy will be carried out by XXX
Purpose of data	<i>Calculation of baseline emissions</i>
Calculation method	<i>Not applicable</i>
Comments	Data will be archived in paper & electronically for a period of 2 years beyond the end of crediting period or of the last issuance of credits for this project activity, whichever occurs later.

5.3 Monitoring Plan

The proposed Monitoring Plan describes the Management System and Procedures to be developed and implemented by XXX for the project XXX in order to ensure consistency with the methodology ACM0002 - Consolidated baseline methodology for grid-connected electricity generation from renewable sources (Version 20.0), which is applied in the context of the project activity.

The developed monitoring plan contains the following sections:

1. Monitoring parameters
2. Monitoring Points
3. Monitoring equipment
4. Management structure
5. Monitoring guidelines, procedures and documentation
6. Regular staff training

All data required for the calculation of emission reductions associated with the project activity will be monitored in accordance with the approved methodology ACM0002 - Consolidated baseline methodology for grid-connected electricity generation from renewable sources (Version 20.0) and will be archived electronically and on paper throughout the crediting period and up to 2 years thereafter.

A Monitoring Management and Quality Assurance System (MMQAS) will be developed and implemented to ensure that the results from the project activity are reliable and transparent.

The complete MMQAS, including data and documents from project operation, will be open for review to the VVBs contracted for verification.

Monitoring Parameters

The tables under section XXX provide information on the required monitoring parameters as per approved methodology ACM0002 - Consolidated baseline methodology for grid-connected electricity generation from renewable sources (Version 20.0) and referred tools, required for the ex-post calculation of project emissions associated with the project activity.

Monitoring Points

Single line diagram

Monitoring equipment

The physical monitoring system consists of the following meter equipment:

Electricity meter: to monitor the quantity of net electricity generation supplied to the grid in year y by the solar plant (EG PJ,y)

Calculations: to calculate Baseline emissions (XXX) by multiplying the quantity of net electricity generation supplied to the grid in year y by the solar plant (EG PJ,y with) with the combined margin CO2 grid emission factor in the year y (EF grid, CM, y)

Management structure

XXX will designate a VCS team to be responsible for carrying out the required monitoring tasks for the project activity. Both management people and technical workers will be present in the VCS team. All of the team must have passed internal monitoring training.

The management structure consists of 3 levels to guarantee a step by step management of the project activity, which will cast concrete responsibilities on the appointed person who will be actually in charge of a specific task.

The 3 levels refer to:

1. Solar Plant Manager
2. VCS project manager

3. VCS staff

Before the start of the crediting period, clear roles and responsibilities will be assigned to all staff involved in the project activity. XXX will appoint a VCS Manager who will be responsible for monitoring activities and data management associated with the project activity. All operators involved in the collection of data and records will be coordinated by the VCS Manager.

Monitoring guidelines, procedures and documentation

The introduced Monitoring Management and Quality Assurance System is building on written guidelines and procedures provided to the operational VCS team staff and to the local VCS management.

The procedures will be developed by the external consultant according to the approved methodology ACM0002 - Consolidated baseline methodology for grid-connected electricity generation from renewable sources (Version 20.0). All tasks under the monitoring plan will be carried out strictly according to the provided guidelines and procedures.

The Monitoring Management and Quality Assurance System documentation is designed to enable VCS staff members to performing the required tasks following the given guidelines and procedures. Providing documentation designed specifically for the required tasks will strongly benefit availability and transparency of data streams and processes which are relevant for the VCS project.

The VCS Project Manager will carry out frequent performance and data cross-checking, to ensure that staff is following the Monitoring Management and Quality Assurance System.

All documents filled out and data acquired during the project will be archived separately for backup.

Regular Staff Training

The Monitoring Management and Quality Assurance System is implemented and will be maintained by an external consultant. In order to qualify local operational staff to carry out required regular tasks external consultant will provide professional training, both in class and practical application, to the designated VCS team staff before the start of the crediting period and ongoing regular training at least once a year for the duration of the project.

Only trained staff will be members of the VCS team and only trained VCS staff members will be allowed to perform any tasks related to monitoring activities.

The VCS Manager is permitted to train new staff. By this it will be ensured that operational VCS staffs are qualified to carry out the relevant tasks under the Monitoring Management and Quality Assurance System.

Based on the annual monitoring results VCS staff will prepare the annual Monitoring Report. An overall AQ/QC on the Monitoring Spreadsheet and Monitoring Report will be carried out by an external Consultant before the documents are submitted to a verifying VVB.

APPENDIX

Use appendices for supporting information. Delete this appendix (title and instructions) where no appendix is required.